

shows the logo of Matplotlib.

After importing Matplotlib, if you execute the command `'matplotlib.__version__'` in the shell, you will be able to find out the version of Matplotlib running on your system. It is not always the case that Python 2 and Python 3 are using the same version of Matplotlib. In my system, Python 2 and IPython use version 1.5.2, whereas Python 3 uses version 2.2.0. Though I have checked the availability of Matplotlib in Python 2, Python 3 and IPython, all the Python programs in this article are tested only with Python 2, so that minor differences between various versions of Matplotlib will not hinder the progress of our discussion. As mentioned earlier, Matplotlib can be used in different contexts—the important ones are within a script, inside a shell, and inside the IPython notebook called Jupyter. An IPython notebook is a browser based interactive data analysis tool that can combine narrative, code, graphics, HTML elements and other multimedia components into a single executable document.



Figure 1: Logo of Matplotlib

Modules in Matplotlib

Pyplot is a module in Matplotlib. It is a collection of command style functions that make Matplotlib work like MATLAB. The functions in Pyplot make some changes to a figure. For example, Pyplot can create a figure and allocate a plotting area for that figure, and then plot a number of lines in the plotting area of the figure.

Another such module provided by Matplotlib is called Pylab. It is a module that bulk imports both `matplotlib.pyplot` and `NumPy`, the Python mathematical package, for easier use. Though it is more convenient to use Pylab, due to this bulk import, its usage is slightly discouraged nowadays. The general rule of thumb proposed by the standard Matplotlib documentation is as follows: Select Pyplot for non-interactive plotting and the Pylab interface for interactive calculations and plotting, as it minimises typing. To simplify matters, our discussion is based solely on Pyplot and we can import `NumPy` separately, if and when required.

Simple line plots in Matplotlib

Let us now look at a simple example involving Matplotlib. The code below shows the Python program `plot1.py`. This, and all the other Python programs discussed in this article, can be downloaded from http://opensourceforu.com/article_source_code/april18/matplot.zip.

```
import matplotlib.pyplot as plt
import numpy as np
a = np.linspace(-10,10,1000)
plt.plot(a,np.sin(a))
plt.plot(a,np.cos(a))
plt.plot(a,np.tan(a))
plt.show()
```

Before executing the program, let us try to understand it in detail. With the first two lines of code, we have imported the packages `matplotlib.pyplot` and `numpy`. The line of code:

```
a = np.linspace(-10,10,1000)
```

...creates a linearly spaced vector 'a' such that it contains 1000 numbers from -10 to 10, with the same common difference between every adjacent pair of numbers. The next three lines of code use the `plot()` function of Pyplot to plot sine, cosine and tangent functions using the vector 'a' already generated. The function `plot()` plots 'y' versus 'x' as lines. In one of the examples that will be discussed later, we will use markers instead of lines to plot, with the `plot()` function. The functions `sin()`, `cos()` and `tan()` are provided by the package `NumPy`. The last line of code:

```
plt.show()
```

...displays the figure on the screen. You can execute the program `plot1.py` in Python 2 with the command:

```
python plot1.py
```

Figure 2 shows the graphics output produced by Matplotlib and Python 2. The same result can also be obtained if you execute every line of code in the shell of Python 2.

Well, it is a bit surprising that the familiar sine and cosine waves are missing in the figure produced. Only the tangent function and two overlapping straight lines are visible in the image. Did we do something wrong? Absolutely not! Then why have we got this particular output? Both sine and cosine functions take values from -1 to 1, whereas the tangent function takes values between $-\infty$ and $+\infty$. So the sine wave and cosine wave look like straight lines due to the large

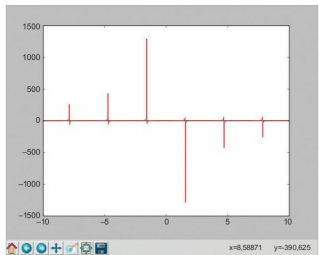


Figure 2: Sine, cosine and tangent functions in Matplotlib